Image Analysis of Electrocardiograms

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Declaration of Authorship

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Image Analysis of Electrocardiograms

Oliver Racey

Abstract

In the medical world, Electrocardiograms are often one of the most common tests used for cardio diagnostics. Many of these ECGs are stored only as printed or scanned images, whilst automated analysis has been developed for digital signals. Converting these signals into readable images can be erroneous - whilst AI models only trained on clean images might not produce the same outputs with real scans and photographs. This project investigates how accurately a deep-learning model can flag possible cardiac abnormalities directly from the ECG image, how well it performs on real-world images, and whether its explanations are reliable enough to support users understand what the

Table of Contents

Declaration of Authorship **i**

Abstract **ii**

Table of Contents **iii**

List of Figures **v**

List of Tables **vi**

Abbreviations **vii**

1 Introduction **1**
1.1 Motivation 1
1.2 Aims and Objectives 1
1.3 Research Question 1

2 Literature Review **2**
2.1 Introduction 2
2.2 Digital Image Processing 2
2.3 Medical Image Processing 3
2.4 ECG Fundamentals 4
2.5 ECG Imaging and Digitisation 6
2.6 Pre-processing and Noise Removal 6
2.7 Feature Extraction 7
2.8 Waveform Detection 7
2.9 AI/ML Algorithms for ECG Analysis 8
2.10 Existing Systems and Limitations 9

3 Methodology **11**
3.1 Introduction 11
3.2 Professional, Legal, Ethical and Social Issues 11

4 Feasibility **12**
4.1 Coding Environment 12
4.2 Design 12

4.3 Prototypes 12
4.4 Datasets 12
4.5 Hardware 12
4.6 Focus Groups 12

A Appendix A **13**

References **14**

List of Figures

2.1 Otsu segmentation of a dermatoscopic image 3
2.2 A single PQRST wave 5
2.3 An ECG paper trace 5

List of Tables

2.1 Key studies in deep learning for ECG classification. 8

Abbreviations (Appearance Order)

ECG	**E**lectro**c**ardio**g**ram
HE	**H**istogram **E**qualization
CLAHE	**C**ontrast **L**imited **A**daptive **H**istogram **E**qualization
HF	**H**omomorphic **F**ilter
WT	**W**avelet **T**ransforms
CNN	**C**onvolutional **N**eural **N**etwork
GN	**G**aussian **N**oise
PDF	**P**robability **D**ensity **F**unction
SNR	**S**ignal-to-**N**oise **R**atio
FT	**F**ourier **T**ransforms
MI	**M**yocardial **I**nfarction
BBB	**B**undle **B**ranch **B**lock
AUC	**A**rea **U**nder the **C**urve
AUROC	**A**rea **U**nder the **R**eceiver **O**perating **C**haracteristic Curve
AMBE	**A**bsolute **M**ean **B**rightness **E**rror
MSE	**M**ean **S**quared **E**rror
LOE	**L**ightness **O**rder **E**rror
PSNR	**P**eak **S**ignal-to-**N**oise **R**atio
SSIM	**S**tructural **S**imilarity of **Im**ages

Chapter 1

Introduction

1.1 Motivation

The electrocardiogram records the electrical activity of the heart over time and is one of the most widely used tests in cardiology. Even though the test is so common, we have only recently been trying to digitise and read them. Most ECGs are stored as raw signals, which does not help interpretation for the majority of the clinicians. On the other hand, ECGs stored as images often have degradations and artefacts that can make them difficult to read. Furthermore, the majority of research on automated ECG analysis has been done on clean or pre-rendered signals, and less on real-world images. This project will investigate how accurately an AI model can flag possible cardiac abnormalities directly from any ECG image, the performance of the model on real-world images, and whether the explanations given are reliable and verbose enough to support clinicians and general users in understanding what the ECG is showing.

1.2 Aims and Objectives

The aims and objectives are as followed:

1. To develop an image preprocessing pipeline that prepares ECG images, either rendered or real, for reliable automated analysis.
2. To develop and evaluate a deep-learning-based, "multi-label" classifier, that flags possible cardiac abnormalities in ECG images, with calibrated confidence scores.
3. To build and evaluate an application that presents flagged abnormalities with visual explanations, as a supporting decision for trained users, and not used as a standalone diagnostic tool.

1.3 Research Question

Chapter 2

Literature Review

2.1 Introduction

The electrocardiogram (**ECG**) is one of the most widely used tests of the heart, so reliable interpretation matters, yet many ECGs are stored or shared as printed or photographed images that software built for digital signals cannot use. This chapter reviews the work needed to build a system that analyses ECGs directly from images, moving from the general to the specific: image processing (Section 2.2) and its use in medical imaging (Section 2.3); the ECG (Section 2.4) and how it is captured as images (Section 2.5); preparing the image, extracting features and detecting waveforms (Sections 2.6–2.8); and the algorithms (Section 2.9) and existing systems (Section 2.10) that define the gaps this project addresses. Sources were found by searching Google Scholar and PubMed between [dates] for terms such as "ECG image classification", "calibration" and "explainable AI ECG", and by following citations from key papers [TODO: replace with your real search log].

2.2 Digital Image Processing

A digital image is a two-dimensional function $f(x, y)$, where x and y are spatial coordinates and the value of f at each pair is the intensity at that point, or pixel [1]. The intensity is the product of the illumination of the scene and the proportion of light its surface reflects [2]:

$$f(x, y) = i(x, y)\, r(x, y),$$

where i is the illumination and r is the reflectance, bounded by 0 (total absorption) and 1 (total reflection). Because i can change whilst r stays the same, one object can look different under different lighting, so one **ECG** trace can form many different images. A computer cannot store a continuous function, so images are digitised by sampling, which divides the image into a grid of pixels, and quantisation, which rounds each value to one of a finite number of intensity levels. More samples give higher resolution, and quantisation is irreversible because rounding discards the difference, albeit at 8 bits per pixel this loss is rarely visible [1].

Enhancement makes an image more suitable for a task, whereas restoration corrects known degradations. **Histogram equalisation (HE)** spreads pixel values across the full intensity

range to increase contrast, but handles local detail poorly; **contrast-limited adaptive histogram equalisation (CLAHE)** works on local regions and limits noise amplification [3]. Neighbourhood filters each have drawbacks: Gaussian filters reduce random noise but blur edges [1], median filters remove isolated specks whilst preserving edges better, and bilateral filtering smooths flat regions whilst keeping boundaries sharp [4]. Frequency-domain methods work on a transformed image. The Fourier transform describes an image as a sum of patterns at different frequencies,

$$f(x,y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u,v)\, e^{j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

so regular repeating patterns, which appear as sharp peaks, can be removed with a notch filter. **Homomorphic filtering (HF)** takes the logarithm of the image, turning the product $i\,r$ into a sum, so that illumination (low frequency) can be suppressed and reflectance (high frequency) boosted; it suits uneven lighting but can amplify noise. **Wavelet transforms (WT)** split an image into components at several scales whilst keeping their position, which suits noise removal [5], and the two can be combined. Removing noise can distort real detail, and no single method suits every image, so the choice depends on the degradation and the task [6].
Segmentation then divides a cleaned image into meaningful regions, using thresholding, which separates foreground from background with an intensity cut-off chosen manually or automatically [7] (Figure 2.1), or edge detection [8]; morphological operations such as opening and closing clean up the result [1].

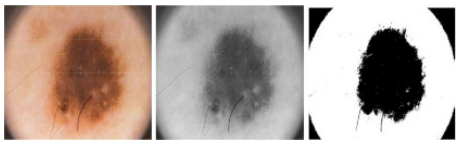


Figure 2.1: Example of Otsu segmentation of a dermatoscopic image, obtained from the PH^2 dataset. Reproduced from [9].

To interpret an image, the most popular approach is a **convolutional neural network (CNN)**, which learns features from labelled examples by applying small filters learned during training [10]. Stacked layers build up from simple edges to complex patterns, so features need not be designed by hand, although a large dataset is needed. Each stage, from cleaning an image to finding its structure to deciding what it shows, trades removing unwanted variation against preserving detail, and these trade-offs matter more when images support medical decisions.

2.3 Medical Image Processing

Medical images follow the same image processing as other types, but must preserve as much anatomical detail as possible. Noise is the main threat: Prasad et al. [11] state that it can degrade diagnostic quality by blurring borders, reducing contrast and obscuring small but

important details. A common model is **Gaussian noise (GN)**, added independently to each pixel,

$$I(x, y) = J(x, y) + N(x, y), \tag{2.1}$$

where I is the noisy image, J is the original and N is drawn from a normal distribution with mean 0 and variance σ^2, so a larger σ^2 means stronger noise.
The general methods of Section 2.2 were adapted for medical use, but each involves a trade-off. Adaptive **HE** was developed for medical image display [12], yet even its contrast-limited form needs the clip limit tuned, and averaging filters suppress noise whilst blurring the edges that carry diagnostic information. Whether an enhancement helps therefore depends on the task: in medical imaging, quality is best defined by how well an observer can perform a specific task, not by how the image looks [13]. A sharper image does not necessarily lead to better decisions, and enhancement that suits a human viewer may not help a classifier.
Because hand-designed methods depend on assumptions about the image, learned methods have largely replaced them; a survey of over 300 studies found **CNN**s used across medical imaging, including cardiac applications [14]. They bring their own problems. Labelled data is expensive and abnormal cases are rare, and performance may not carry over to new data: Zech et al. [15] found that networks for detecting pneumonia in chest radiographs performed differently across three hospital systems. Models may also rely on shortcuts that correlate with the label instead of the features that matter [16], and because predictions may influence patient decisions, systems must explain their output, for example with Grad-CAM [17], and report calibrated confidence [18]. These points apply directly to this project: preprocessing will be judged by its effect on classification and not by appearance, and the gap between rendered training images and real test images is a form of the generalisation problem, with the added risk of rendering-specific shortcuts.

2.4 ECG Fundamentals

The **ECG** is an electrical tracing of the heart, recorded non-invasively from the surface of the body [19]. It shows how the impulse that triggers each heartbeat spreads through the heart, so it is well suited to detecting problems of rhythm and conduction. Each heartbeat produces a repeating sequence of deflections (Figure 2.2): the P wave reflects activation of the atria, the QRS complex activation of the ventricles, and the T wave their recovery. The intervals between these features carry diagnostic information: the PR interval reflects the delay as the impulse passes from the atria to the ventricles, the QRS duration how quickly the ventricles are activated, and the QT interval spans ventricular activation and recovery. Abnormalities are defined by changes to these features, so a model working from an image must find them and judge their shape and spacing.
A standard resting **ECG** records twelve leads, six limb leads and six chest leads [19]. Each lead views the same heartbeat from a different direction, so a problem in one region of the heart often appears in only some leads. A page cannot fit twelve full-length traces, so the printout cuts each into a short piece and arranges them on a 3×4 grid [21] (Figure 2.3); a single image therefore contains twelve traces, and a classifier must learn which leads matter for which condition. On the grid, the distance x along the paper and the deflection y convert to time and voltage by $t = x/v$ and $V = y/g$, where v is the paper speed and g the gain

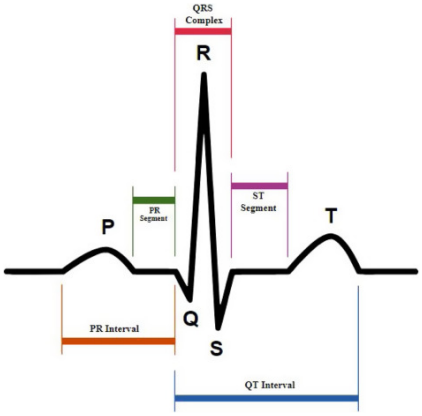


Figure 2.2: A single heartbeat, showing the P wave, QRS complex, T wave and main intervals. Generated by the author from [20].

(25 mm/s and 10 mm/mV as standard [19]). Clinicians read rate and intervals by counting squares; for example, the heart rate follows from the RR interval,

$$\mathrm{HR} = \frac{60}{\mathrm{RR}} = \frac{300}{N},$$

where RR is the time in seconds between successive R peaks and N is the number of large squares between them. Timing and amplitude can thus only be read from an image if the grid calibration is known, so skewing, cropping, resizing or photographing the page all distort it.

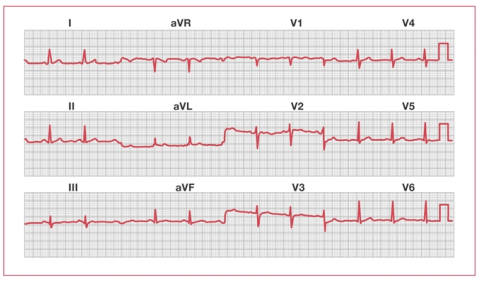


Figure 2.3: A single trace, showing a 3×4 grid of leads [21].

Clinically important abnormalities can be grouped by what they affect. Abnormal heart rates and rhythms are known as arrhythmias [22]; they include atrial fibrillation, where beats are irregularly spaced and clear P waves are absent [23], and atrial flutter, which shows a regular saw-tooth pattern in the inferior leads [24]. Conduction disturbances such as **bundle branch blocks (BBB)** usually widen the QRS complex [25], **myocardial infarction (MI)** and other ischaemic changes often alter the ST segment and T wave in the leads facing the affected region, and hypertrophy of the heart's chambers is associated with changes in amplitude [26]. Some of these signs depend on the shape of the trace, whereas others depend on timing and regularity, which in an image depends on the grid scale.

Public datasets encode these findings as labels. PTB-XL contains 21,837 ten-second twelve-lead **ECG**s from about 19,000 patients, annotated by cardiologists, with diagnostic labels grouped into five superclasses (normal, **MI**, ST/T change, conduction disturbance and hypertrophy) plus form and rhythm statements, and a single record can carry several labels [27]. It provides signals, not images; a dataset from UMT provides real ECG images instead [28]. These fundamentals define the classes this project will detect and the two kinds of information the classifier must use: the shape of each beat, and the timing between beats.

2.5 ECG Imaging and Digitisation

Although ECGs are recorded electronically, they are frequently printed, scanned or photographed, and such images cannot be used directly by software that expects time-series data [29]. They also vary widely: the ECG-Image-Database, for example, contains printed ECGs that were scanned or photographed under different lighting conditions and camera angles, some deliberately soaked, stained or left to grow mould, giving shadows, perspective shifts, blur, creases and stains [30]. This is the illumination and reflectance effect of Section 2.2. Image resolution also limits how precisely the trace can be read, so it directly limits the timing measurements, such as intervals and rate, that some classes depend on. Two approaches address the problem. Digitisation reconstructs the time-series signal from the image so that established signal-based methods can be applied. This gives measurable intervals, but needs several steps (locating the grid, separating the leads and tracing each waveform), and an error at any step carries through to the reconstructed signal. The George B. Moody PhysioNet Challenge 2024, for example, evaluated digitisation by the **signal-to-noise ratio (SNR)** of the recovered signal [31]. The alternative is to classify the image directly (Section 2.9), which avoids reconstruction, but the model must then learn timing from pixels, so its results depend on how the images were produced, which is the generalisation problem of Section 2.3. ECG-Image-Kit generates synthetic ECG images from real signals, with added artefacts [29], and the ECG-Image-Database provides 35,595 images of 1,977 recordings from PTB-XL and Emory Healthcare that were printed, degraded, and then scanned or photographed [30]. This project analyses ECG images directly and avoids a full digitisation step: it trains on images rendered from PTB-XL and tests on real-world images, so that the effect of real imaging conditions can be measured.

2.6 Pre-processing and Noise Removal

Preprocessing corrects problems introduced during printing and capture, using the methods of Section 2.2 matched to problems that occur in **ECG** images, and each carries a risk. Uneven lighting and shadows can be reduced with **CLAHE** [3], [12] or **HF** [2], at the cost of amplifying noise. Noise and specks (Equation 2.1) can be reduced with median filtering, bilateral filtering [32] or non-local means [33], although smoothing can flatten thin peaks. The printed grid can be removed by colour segmentation or notch filtering [1], which struggles with faded or monochrome prints, and rotation can be corrected by line detection, which cannot fully undo the perspective distortion of a photograph.
The grid needs particular care because it is both noise and information: it crosses the trace and can confuse a classifier, but it is also the only record of the time and voltage scale (Sec-

tion 2.4), so removing it without recording the scale first would discard what the measurements depend on. The trace is also thin, so smoothing or enhancement can remove P waves and sharp QRS peaks that are only a few pixels wide. It is not certain that preprocessing helps a **CNN** at all, since a network trained on varied, degraded images may learn to cope with them. Following the task-based view of Section 2.3, preprocessing is therefore a design choice, judged by its effect on classification performance and not by how the image looks.

2.7 Feature Extraction

Before deep learning, automated **ECG** analysis relied on hand-designed features: once the beats were located, quantities such as interval lengths and wave amplitudes were measured and passed to a classical classifier, as in the heartbeat classification of Chazal et al. [34], which combined morphology and interval features. Such features have a clear clinical meaning, which makes the model easier to explain, and timing features are especially useful for rhythm: the regularity of the RR intervals, for example, separates a regular rhythm from an irregular one such as atrial fibrillation. The alternative is learned features: a **CNN** builds its features directly from pixels [35], so no one has to decide in advance what to measure. Hand-designed features are interpretable but depend on every earlier step succeeding, since a missed beat or a broken trace corrupts every feature that follows; learned features avoid this dependency but are harder to explain and need a large labelled dataset.
This project applies both. A **CNN** learns shape-based features from the image, which suits the set A classes. Classes defined by timing, such as heart rate and atrial fibrillation, depend on the spacing between beats, which a model can only judge correctly if the scale is consistent across images; measured timing features from the waveform-detection step (Section 2.8) address this.

2.8 Waveform Detection

Waveform detection finds the parts of the **ECG** that features are measured from. In digital signals this usually means locating the QRS complex: the Pan–Tompkins algorithm does so in real time by band-pass filtering and differentiating the signal, squaring it, smoothing it with a moving window, and applying adaptive thresholds [36]. For an image, the first task is finding the trace itself. Published pipelines binarise the image, remove the grid and text, repair gaps in the trace and locate each lead, using a Hough transform to find the grid lines or lead baselines [37]. Each column of the isolated trace is then reduced to one value, for example the median vertical position of the trace pixels [37], and converted to time and voltage using the grid scale, after which peaks can be detected as in the signal case.
Explicit detection gives measurable intervals and a rate that a clinician can check, and it can be accurate: Randazzo et al. [38] reported an average correlation of 0.94 with reference signals on images from 16 subjects, and Wu et al. [37] about 99% on standard layouts once images with overlapping leads were excluded. It is also fragile: the correlation of Wu et al. [37] fell to about 96% on printed and rescanned **ECGs**, whilst a different pipeline scored a signal-to-noise ratio of $-2.44\,\mathrm{dB}$ on 100 rendered PTB-XL images, a poor reconstruction by that measure. A missed or false peak changes the RR intervals and every timing feature that follows, so errors here carry through to the prediction. This project therefore applies

waveform detection to locate the trace and the beats and supply timing features alongside the learned image features, evaluating the detection step on real-world images and not only on rendered ones.

2.9 AI/ML Algorithms for ECG Analysis

Deep learning has largely replaced feature-based methods for **ECG**classification (Table 2.1). On signals, networks reached cardiologist-level performance on single-lead rhythm classification [39], outperformed cardiology residents on 12-lead abnormalities [40], and a PTB-XL benchmark found convolutional architectures, especially ResNet- and Inception-based ones, performed best [41]. Image-based work is newer. Sangha et al. [42] trained a **CNN** on images generated from signals and tested it on held-out, external and printed ECGs, Gliner et al. [43] used domain adaptation to transfer a model to phone-captured images, and Wang et al. [44] fine-tuned a ConvNeXt backbone first on synthetic and external images. Because labelled **ECG** images are limited, such work relies on transfer learning, in which networks such as ResNet [45], pretrained on natural images, learn general features that transfer to new tasks [46].

Table 2.1: Key studies in deep learning for ECG classification.

Study	**Input and data**	**Reported result**	**Limitation for this project**
Hannun et al. [39]	Single-lead signals; 91,232 ECGs, 53,549 patients	12 rhythm classes; F1 0.837 vs. 0.780 for cardiologists	Single lead; signals, not images
Ribeiro et al. [40]	12-lead signals; over two million exams	Six abnormalities; F1 above 80%, specificity above 99%	Signals, not images
Strodthoff et al. [41]	12-lead signals, PTB-XL	Convolutional models performed best	Signals; hidden weakness in subclasses
Sangha et al. [42]	Images generated from signals; printed ECGs to test	Strong performance on held-out, external and printed ECGs [TODO: AUROC]	Training images generated from signals [TODO: check]
Gliner et al. [43]	79,226 scanned labelled ECGs plus unlabelled phone-captured frames	Average ROC-AUC 0.91 on phone-captured images	Phone images made by filming printed pages; six unseen formats tested
Wang et al. [44]	Synthetic PTB-XL images plus external scanned sets	AUROC about 0.97 on the hidden test set	Synthetic test data; possible leakage from PTB-XL

These results do not transfer directly to images. Hannun et al. [39] used a single lead, with test labels from a consensus committee of cardiologists, and both they and Ribeiro et al. [40] worked on signals, which arrive with a known sampling rate and amplitude scale; an image carries its scale only through the printed grid, so timing must be recovered from pixels (Section 2.5). Label quality is a further limit. PTB-XL's labels come from at most two cardiologists [27], and agreement between cardiologists can be only poor to moderate [47], so a high AUROC partly measures agreement with imperfect labels. Studies also use different datasets, label sets and metrics (F1, specificity, AUROC), so their headline numbers cannot be ranked directly.

Evaluation therefore needs particular care. ECG datasets are imbalanced, so accuracy can be high even when rare but important conditions are missed; measures such as sensitivity, specificity and the Matthews correlation coefficient give a more truthful picture [48], and data should be split by patient, for which PTB-XL provides suggested folds [27]. A model used for decision support must also show how far its outputs can be trusted. Modern neural networks are often overconfident, but temperature scaling can calibrate their scores [18]. For multi-label classification each class has its own sigmoid output, so each class score z_k is divided by a temperature T fitted on a held-out validation set,

$$\hat{p}_k = \sigma\left(\frac{z_k}{T}\right), \tag{2.2}$$

by minimising the binary cross-entropy of the scaled outputs. Because dividing by T does not change the ranking of scores, AUC is unaffected and only the reliability of the probabilities changes, which is measured with the expected calibration error for each class. Guo et al. described the method for single-label classification, so applying Equation (2.2) to independent sigmoid outputs is an adaptation. Grad-CAM heatmaps show which image regions influenced a prediction [17], but not why a clinician would reach the same conclusion, so they need checking against known cases.

2.10 Existing Systems and Limitations

Automated interpretation has been part of clinical ECG machines for decades: millions of ECGs are recorded each year, and most are analysed by computer with an immediate report [49]. However, limitations in their diagnostic accuracy were recognised early and persist, and the authors state that over-reading and confirmation by an experienced ECG reader are essential, because inexperienced physicians may accept an automated reading without question [49]. These systems also work on the digital signal at the point of recording, so they cannot analyse an ECG that exists only as a printout or image.

Deep learning systems have achieved strong results (Section 2.9), but four gaps remain that are relevant to this project. First, image-based classification has mostly been tested on synthetic or limited real images. Gliner et al. [43] adapted a model to phone-captured ECGs, but their images were produced by filming printed pages and only six unseen formats were tested, and a challenge-winning pipeline was evaluated on synthetic images rendered from PTB-XL, which its authors note may inflate results through data leakage [44]. Second, calibration has been studied for signal-based atrial fibrillation detection, where uncertainty estimates improved the expected calibration error [50], but the image-based studies above report discrimination measures such as AUROC. Third, explanations are seldom assessed: a review of explainable AI in cardiology found that 43% of the studies examined did not evaluate their explanations at all, and only 11% involved clinicians [51]. Fourth, how results are presented matters and depends on the user. In interviews with 23 clinicians, accuracy mattered more than explanation, and visualisations of what the algorithm analysed were preferred [52], while a simulated-AI experiment with 44 cardiologists found that explanations raised the trust of novices more than experts and could mislead [53]. Datasets also reflect limited populations and recording conditions: PTB-XL, for instance, was collected between 1989 and 1996 using devices from a single manufacturer [27], so models may not generalise

to other settings, an ethical concern discussed in the Professional, Legal, Ethical and Social Issues section.
These gaps define the aims of this project. A preprocessing and detection pipeline tested on real-world images addresses the first gap; a calibrated multi-label classifier evaluated with imbalance-aware metrics addresses the second; and an application that presents explained results to different users addresses the third and fourth.

Chapter 3

Methodology

3.1 Introduction

3.2 Professional, Legal, Ethical and Social Issues

Chapter 4

Feasibility

4.1 Coding Environment

4.2 Design

4.3 Prototypes

4.4 Datasets

4.5 Hardware

4.6 Focus Groups

Appendix A

Appendix A

Scripts

Supplementary material...

GitHub Repository

Project planning is core aspect of any dissertation (especially for the initial groundwork). Key tool is Git/Github

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